Hai Jiang

CV Engineer

AI / ML Engineer
Imaging Algorithms

+86 198 6771 3757

Guangdong, China

jianghai97@mail.nankai.edu.cn

pigejianghai

Profile

Computer vision and imaging algorithm engineer with five years of experience in medical imaging, industrial AI, segmentation, detection, OCR, geometric feature extraction, and deployment optimization. M.Sc. in Computational Mathematics from Nankai University. First-author ISBI 2025 paper and Medical Physics manuscript under revision. Strong in PyTorch, C++, ONNX, TensorRT, and Linux inference pipelines, with a focus on robust, efficient perception systems for Apple products.

Professional Experience

2026.04 – Present
Image Algorithm Engineer
Shenzhen Image Technology Co., Ltd.

- Develop C++ geometry algorithms for PCB copper-layer Gerber parsing and polygon feature extraction, converting raw contour data into structured inspection-ready representations.
- Extract edges, traces, corners, blocks, pads, and other manufacturing-critical geometric entities from large-scale polygon sets for downstream process inspection and quality analysis.
- Design efficient computational geometry and topology analysis modules, optimize memory usage and throughput, and maintain unit tests and algorithm documentation for production use.

2025.08 – 2026.04
Computer Vision Algorithm Engineer
Shenzhen Zhongrui Power Technology Co., Ltd.

- Built an automated AI engine for a property-management data platform using YOLO, enabling object recognition, segmentation, and classification from user-uploaded images.
- Integrated OCR for receipts, signs, and document-like scenes, transforming detected text into structured outputs for platform workflows.
- Packaged the inference pipeline behind standardized APIs for web integration, using ONNX export and TensorRT acceleration to improve real-time response latency.

2022.07 – 2025.07
Research Assistant
Computational Medical Imaging Laboratory, Sun Yat-sen University

- Designed and validated algorithms for national medical AI projects, covering MRI, dual-energy CT, and digital breast tomosynthesis; contributed to grant proposals, final reports, and industry collaboration.
- Proposed an anatomy-guided multitask learning framework with attention for placenta accreta spectrum classification on MRI, achieving AUC $>$ 0.99 and first-author acceptance at ISBI 2025.
- Built a multi-channel CNN with a virtual-class strategy for lymph-node metastasis prediction from dual-energy CT, improving the average AUC to 0.85; manuscript under revision at Medical Physics.
- Contributed to multi-view CNN and Siamese-network research for atypical architectural distortion detection in DBT, supporting a Guangdong Science and Technology Progress Award submission.

Education

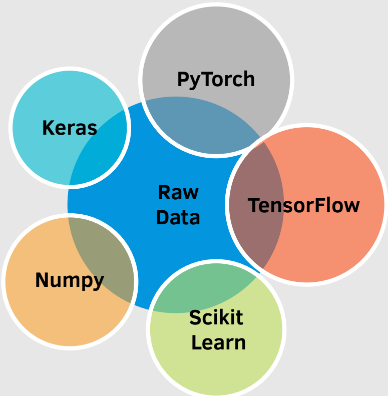
2019.09 – 2022.06
M.Sc. in Computational Mathematics
Nankai University Project 985 / 211 / Double First-Class

- Research: optimization algorithms (Primal-Dual Learning, ADMM), deep learning, and MRI reconstruction.
- Thesis: Deep Unfolding Networks for MRI Image Reconstruction.

2015.09 – 2019.06
B.Eng. in Information Security
Lanzhou University Project 985 / 211 / Double First-Class

- Core coursework: data structures, algorithm design and analysis, machine learning, and cryptography.

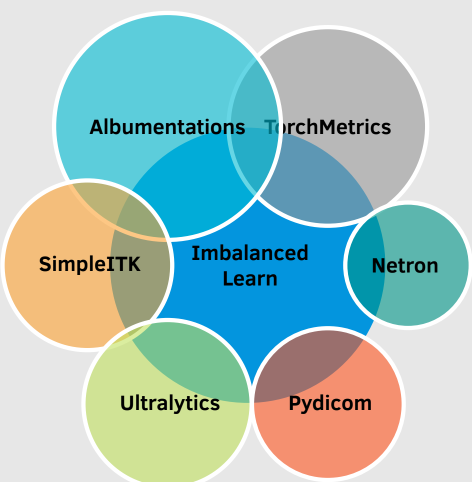
ML Libraries



Engineering Skills

- Python
- PyTorch • TensorFlow
- ONNX • TensorRT
- Linux • Git • CMake
- C++ • C • MATLAB

Python Libraries



Languages

Mandarin Chinese (Native)

English (C1, IELTS 7.0)

Selected Publications & Research Outcomes

ISBI 2025 — *Anatomy-Guided Multitask Learning for Placenta Accreta Spectrum and Its Subtypes*

Accepted — First author. Developed an anatomy-guided multitask learning framework for MRI-based PAS classification, reaching AUC > 0.99.

Medical Physics — *DECT-based Space-squeeze Method for Multi-class Classification of Metastasis Lymph Nodes in Breast Cancer*

Revision — First author. Designed a multi-channel CNN and virtual-class strategy to fuse multi-energy CT information, reaching average AUC = 0.85.

PMB — *Atypical Architectural Disorder Detection in Digital Breast Tomosynthesis: a multi-view computer-aided detection model with ipsilateral learning*

Accepted — Co-author. Introduced Triplet Loss and Siamese learning to improve abnormal-region detection from multi-view DBT.

Computer Vision & Imaging Projects

RBC System Cell AI — **TensorFlow-Based Red Blood Cell Classification System**

- Designed a six-class medical image classifier for red blood cells, addressing class imbalance, overfitting, and real-time inference constraints.
- Applied sampling strategies, weighted losses, custom lightweight networks, and pooling-operator replacement to improve minority-class accuracy and generalization.

PAS MRI Multitask — **Multitask Learning with Anatomical Priors**

- Used nnU-Net to segment placenta and uterine myometrium from 414 T2-weighted MRI cases, achieving mean Dice = 0.92.
- Built a multitask model for binary and subtype classification, fusing anatomical priors via soft attention; binary AUC > 0.99, improving the baseline by 20%.

DECT Breast CA — **Multi-Energy Fusion and Virtual-Class Strategy**

- Built a multi-channel CNN on 227 dual-energy CT cases across 11 energy levels (40–140 keV), using channel attention to strengthen feature fusion.
- Proposed a virtual-class strategy to reduce class overlap and improve average AUC to 0.85, a 10% gain over the baseline.

DBT Detection — **Multi-View Detection with Siamese Learning**

- Developed multi-view CNN components for atypical architectural distortion detection in digital breast tomosynthesis.
- Combined Triplet Loss and Siamese learning to improve abnormal-region representation and cross-view consistency.

MRI Recon Deep Unfolding — **Deep ADMM-Net Implementation and Optimization**

- Implemented a Deep Unfolding reconstruction network based on ADMM optimization for end-to-end MRI reconstruction.
- Used PyTorch and C++ to improve reconstruction quality under undersampling, grounded in compressed sensing and ADMM analysis.